

AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 2: Differential Calculus

Adam Prugel-Bennett

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1 Keywords

- Differentiation, linearity, product rule, chain rule, inverse functions, gradients, Jacobians, matrix derivatives, traces

2 Main Points

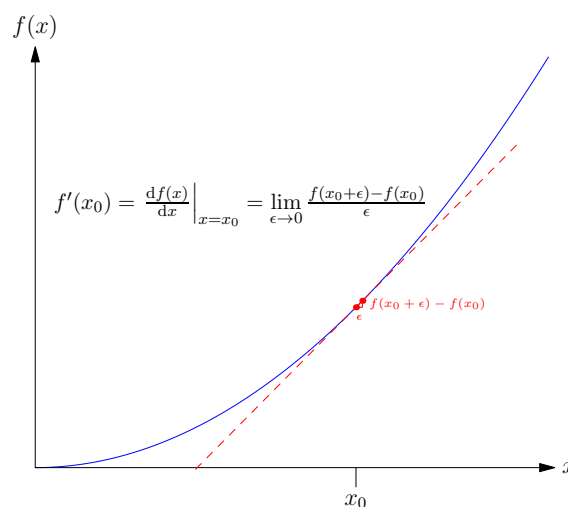
2.1 Why Calculus?

- Differentiation is the fundamental tool of optimisation, and optimisation is how learning machines are trained
- Integration is how we take expectations over continuous distributions
- You met the rules at A-level, but they are easy to forget, and we will use some quite sophisticated tricks. It pays to understand where the tricks come from

2.2 Differentiation from First Principles

- The derivative is the limit of the slope of a chord,

$$f'(x) = \frac{df(x)}{dx} = \lim_{\epsilon \rightarrow 0} \frac{f(x + \epsilon) - f(x)}{\epsilon}$$



- An equivalent statement, used over and over again in this course, is

$$f(x + \epsilon) = f(x) + \epsilon f'(x) + O(\epsilon^2)$$

To find a derivative, expand $f(x + \epsilon)$ and read off the coefficient of ϵ

- **Polynomials.**

$$\frac{dx^2}{dx} = \lim_{\epsilon \rightarrow 0} \frac{(x^2 + 2\epsilon x + \epsilon^2) - x^2}{\epsilon} = \lim_{\epsilon \rightarrow 0} (2x + \epsilon) = 2x$$

More generally $(x + \epsilon)^n = x^n + n\epsilon x^{n-1} + O(\epsilon^2)$, so $dx^n/dx = nx^{n-1}$

- **Linearity.** Splitting the limit,

$$\frac{d}{dx} (a f(x) + b g(x)) = a f'(x) + b g'(x)$$

Differentiation is a linear operation

2.3 Product and Chain Rules

- **Product rule.** Multiplying out the expansions of $f(x + \epsilon)$ and $g(x + \epsilon)$,

$$f(x + \epsilon) g(x + \epsilon) = f(x) g(x) + \epsilon (f'(x) g(x) + f(x) g'(x)) + O(\epsilon^2)$$

so the derivative of $h(x) = f(x) g(x)$ is $h'(x) = f'(x) g(x) + f(x) g'(x)$

- **Chain rule.** For $h(x) = f(g(x))$,

$$h(x + \epsilon) = f(g(x) + \epsilon g'(x) + O(\epsilon^2)) = f(g(x)) + \epsilon g'(x) f'(g(x)) + O(\epsilon^2)$$

so $h'(x) = g'(x) f'(g(x))$

- Together the product and chain rules let us differentiate almost anything
- The chain rule is often easier to remember as

$$\frac{df(g(x))}{dx} = \frac{df(g)}{dg} \frac{dg(x)}{dx}$$

For example

$$\frac{d e^{\cos(x^2)}}{dx} = \frac{d e^{\cos(x^2)}}{d \cos(x^2)} \frac{d \cos(x^2)}{dx^2} \frac{dx^2}{dx} = e^{\cos(x^2)} (-\sin(x^2)) 2x = -2x \sin(x^2) e^{\cos(x^2)}$$

2.4 Inverse Functions

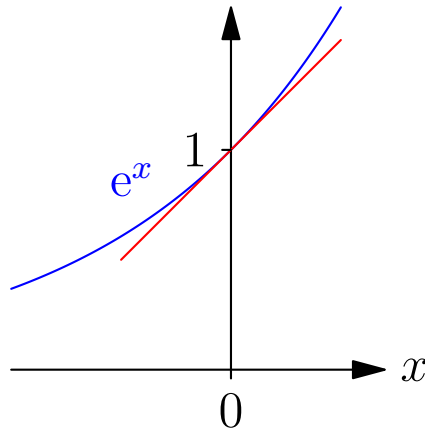
- If $g = f^{-1}$, so $g(f(x)) = x$, differentiating both sides with the chain rule gives $f'(x) g'(f(x)) = 1$
- Writing $y = f(x)$,

$$\frac{df^{-1}(y)}{dy} = \frac{1}{f'(f^{-1}(y))}$$

2.5 Exponentials and Logarithms

- Since $e^\epsilon = 1 + \epsilon + O(\epsilon^2)$ and $e^{x+\epsilon} = e^x e^\epsilon$,

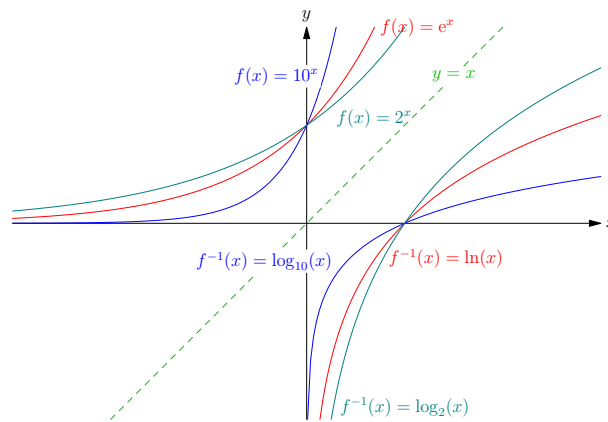
$$e^{x+\epsilon} = e^x + \epsilon e^x + O(\epsilon^2) \quad \implies \quad \frac{d e^x}{dx} = e^x$$



- By the chain rule $d e^{c x} / d x = c e^{c x}$, and more generally $d e^{g(x)} / d x = g'(x) e^{g(x)}$
- For any base, $a^x = e^{x \ln(a)}$, so $d a^x / d x = \ln(a) a^x$
- The natural logarithm is the inverse of the exponential, $\ln(e^x) = x$ and $e^{\ln(y)} = y$. The inverse function rule with $f(x) = e^x$ gives

$$\frac{d \ln(y)}{d y} = \frac{1}{e^{\ln(y)}} = \frac{1}{y}$$

- The lecture animation shows e^x and $\ln(x)$ as reflections of one another in the line $y = x$: reflecting a curve swaps rise and run, which is why the slopes of inverse functions are reciprocals



2.6 Derivatives in High Dimensions

- A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ changes at different rates in different directions. The derivative in a unit direction $\mathbf{u}$ is the *directional derivative*

$$\partial_{\mathbf{u}} f(\mathbf{x}) = \lim_{\epsilon \rightarrow 0} \frac{f(\mathbf{x} + \epsilon \mathbf{u}) - f(\mathbf{x})}{\epsilon}$$

- Taking $\mathbf{u} = \delta_i$, the unit vector along the i^{th} axis, gives the partial derivative $\partial f(\mathbf{x}) / \partial x_i$
- Expanding to first order,

$$f(\mathbf{x} + \epsilon \mathbf{u}) = f(\mathbf{x}) + \epsilon \mathbf{u}^T \nabla f(\mathbf{x}) + O(\epsilon^2)$$

where the *gradient* $\nabla f(\mathbf{x})$ is the column vector of partial derivatives. This is the start of the high dimensional Taylor expansion

2.7 Computing Gradients

- **Component-wise.** Writing $\mathbf{w}^\top \mathbf{M} \mathbf{w} = \sum_{i,j} w_i M_{ij} w_j$ and differentiating with respect to w_k picks out the terms with $i = k$ or $j = k$:

$$\frac{\partial}{\partial w_k} \sum_{i,j} w_i M_{ij} w_j = \sum_j M_{kj} w_j + \sum_i w_i M_{ik}$$

so $\nabla \mathbf{w}^\top \mathbf{M} \mathbf{w} = (\mathbf{M} + \mathbf{M}^\top) \mathbf{w}$. Tedious, but when you are confused, go back to components

- **By expansion.** A slicker method is to expand $f(\mathbf{x} + \epsilon \mathbf{u})$. For $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{M} \mathbf{x} + \mathbf{a}^\top \mathbf{x}$,

$$\begin{aligned} f(\mathbf{x} + \epsilon \mathbf{u}) &= f(\mathbf{x}) + \epsilon (\mathbf{u}^\top \mathbf{M} \mathbf{x} + \mathbf{x}^\top \mathbf{M} \mathbf{u} + \mathbf{a}^\top \mathbf{u}) + O(\epsilon^2) \\ &= f(\mathbf{x}) + \epsilon \mathbf{u}^\top (\mathbf{M} \mathbf{x} + \mathbf{M}^\top \mathbf{x} + \mathbf{a}) + O(\epsilon^2) \end{aligned}$$

using $\mathbf{x}^\top \mathbf{M} \mathbf{u} = \mathbf{u}^\top \mathbf{M}^\top \mathbf{x}$ and $\mathbf{a}^\top \mathbf{u} = \mathbf{u}^\top \mathbf{a}$. Reading off the coefficient, $\nabla f(\mathbf{x}) = \mathbf{M} \mathbf{x} + \mathbf{M}^\top \mathbf{x} + \mathbf{a}$

2.8 Jacobians and the Chain Rule in High Dimensions

- Consider a composition $h(\mathbf{x}) = f(\mathbf{g}(\mathbf{x}))$ with $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $f : \mathbb{R}^m \rightarrow \mathbb{R}$. The vector valued $\mathbf{g}$ is just m scalar functions $g_1, \dots, g_m$

- Perturbing each one,

$$\mathbf{g}(\mathbf{x} + \epsilon \mathbf{u}) = \mathbf{g}(\mathbf{x}) + \epsilon \mathbf{J} \mathbf{u} + O(\epsilon^2)$$

where the *Jacobian* $\mathbf{J}$ is the $m \times n$ matrix with $J_{ij} = \partial g_i(\mathbf{x}) / \partial x_j$

- Then, to first order,

$$h(\mathbf{x} + \epsilon \mathbf{u}) = f(\mathbf{g}(\mathbf{x}) + \epsilon \mathbf{J} \mathbf{u}) = h(\mathbf{x}) + \epsilon \mathbf{u}^\top \mathbf{J}^\top \nabla_{\mathbf{g}} f(\mathbf{g}(\mathbf{x}))$$

- Comparing with $h(\mathbf{x} + \epsilon \mathbf{u}) = h(\mathbf{x}) + \epsilon \mathbf{u}^\top \nabla_{\mathbf{x}} h(\mathbf{x})$ gives the chain rule in high dimensions,

$$\nabla_{\mathbf{x}} h(\mathbf{x}) = \mathbf{J}^\top \nabla_{\mathbf{g}} f(\mathbf{g}(\mathbf{x})) \quad \text{or} \quad \frac{\partial h(\mathbf{x})}{\partial x_i} = \sum_{j=1}^m \frac{\partial f(\mathbf{g})}{\partial g_j} \frac{\partial g_j(\mathbf{x})}{\partial x_i}$$

- Backpropagation in a neural network is this rule applied layer by layer: the gradient is passed backwards by multiplying by transposed Jacobians

2.9 Differentiating with Respect to Matrices

- Losses often depend on a matrix of weights, e.g. $L(\mathbf{W}) = (\mathbf{a}^\top \mathbf{W} \mathbf{b} - c)^2$. A minimum is a point $\mathbf{W}^*$ where, for every matrix $\mathbf{U}$, $L(\mathbf{W}^* + \epsilon \mathbf{U}) - L(\mathbf{W}^*) = O(\epsilon^2)$
- Define $\partial L(\mathbf{W}) / \partial \mathbf{W}$ as the matrix of partial derivatives, with elements $\partial L / \partial W_{ij}$. The same argument as for vectors gives

$$L(\mathbf{W} + \epsilon \mathbf{U}) = L(\mathbf{W}) + \epsilon \operatorname{tr} \left(\mathbf{U}^\top \frac{\partial L(\mathbf{W})}{\partial \mathbf{W}} \right) + O(\epsilon^2)$$

where $\operatorname{tr}(\mathbf{U}^\top \mathbf{G}) = \sum_{ij} U_{ij} G_{ij}$ is the inner product between matrices

- **Example.** For $L(\mathbf{W}) = (\mathbf{a}^\top \mathbf{W} \mathbf{b} - c)^2$,

$$L(\mathbf{W} + \epsilon \mathbf{U}) = L(\mathbf{W}) + 2\epsilon (\mathbf{a}^\top \mathbf{W} \mathbf{b} - c) \mathbf{a}^\top \mathbf{U} \mathbf{b} + O(\epsilon^2)$$

and $\mathbf{a}^\top \mathbf{U} \mathbf{b} = \sum_{ij} a_i U_{ij} b_j = \operatorname{tr}(\mathbf{U}^\top \mathbf{a} \mathbf{b}^\top)$, so $\partial L / \partial \mathbf{W} = 2 (\mathbf{a}^\top \mathbf{W} \mathbf{b} - c) \mathbf{a} \mathbf{b}^\top$

2.10 Traces

- The trace is the sum of the diagonal elements, $\text{tr } \mathbf{A} = \sum_i A_{ii} = \text{tr } \mathbf{A}^\top$
- It is linear, $\text{tr}(c \mathbf{A}) = c \text{tr } \mathbf{A}$ and $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr } \mathbf{A} + \text{tr } \mathbf{B}$
- $\text{tr}(\mathbf{A} \mathbf{B}) = \sum_{ij} A_{ij} B_{ji} = \text{tr}(\mathbf{B} \mathbf{A})$, so a trace is unchanged by cycling the matrices in a product:

$$\text{tr}(\mathbf{A} \mathbf{B} \mathbf{C} \mathbf{D}) = \text{tr}(\mathbf{D} \mathbf{A} \mathbf{B} \mathbf{C}) = \text{tr}(\mathbf{C} \mathbf{D} \mathbf{A} \mathbf{B}) = \text{tr}(\mathbf{B} \mathbf{C} \mathbf{D} \mathbf{A})$$

- **Quick matrix differentiation.** Since $\partial_{\mathbf{U}} f(\mathbf{X}) = \text{tr}(\mathbf{U}^\top \partial f(\mathbf{X}) / \partial \mathbf{X})$, expand, then use the trace rules to move $\mathbf{U}^\top$ to the front. For example

$$\partial_{\mathbf{U}} \text{tr}(\mathbf{A} \mathbf{X} \mathbf{B}) = \text{tr}(\mathbf{A} \mathbf{U} \mathbf{B}) = \text{tr}(\mathbf{B}^\top \mathbf{U}^\top \mathbf{A}^\top) = \text{tr}(\mathbf{U}^\top \mathbf{A}^\top \mathbf{B}^\top)$$

$$\text{so } \partial \text{tr}(\mathbf{A} \mathbf{X} \mathbf{B}) / \partial \mathbf{X} = \mathbf{A}^\top \mathbf{B}^\top$$

- Many more useful results are collected in *The Matrix Cookbook*

3 Exercises

3.1 First principles

- Show from the definition that $d(1/x)/dx = -1/x^2$

3.2 Chain rule

- Differentiate x^x (hint: write it as an exponential)

3.3 Inverse functions

- Use the inverse function rule to find $d \arcsin(y)/dy$

3.4 The logistic function

- Show that $\sigma(x) = 1/(1 + e^{-x})$ satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x))$

3.5 Least squares

- By expanding $f(\mathbf{w} + \epsilon \mathbf{u})$, find the gradient of $f(\mathbf{w}) = \|\mathbf{y} - \mathbf{X} \mathbf{w}\|^2$

3.6 A matrix derivative

- Find $\partial L / \partial \mathbf{W}$ for $L(\mathbf{W}) = \|\mathbf{W} \mathbf{x} - \mathbf{y}\|^2$

4 Answers

4.1 First principles

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$$\frac{1}{\epsilon} \left(\frac{1}{x + \epsilon} - \frac{1}{x} \right) = \frac{1}{\epsilon} \frac{x - (x + \epsilon)}{x(x + \epsilon)} = -\frac{1}{x(x + \epsilon)} \rightarrow -\frac{1}{x^2}$$

4.2 Chain rule

- $x^x = e^{x \ln(x)}$, so by the chain and product rules

$$\frac{dx^x}{dx} = (\ln(x) + 1) e^{x \ln(x)} = (\ln(x) + 1) x^x$$

4.3 Inverse functions

- With $f = \sin$ and $f' = \cos$,

$$\frac{d \arcsin(y)}{dy} = \frac{1}{\cos(\arcsin(y))} = \frac{1}{\sqrt{1-y^2}}$$

using $\cos(\theta) = \sqrt{1 - \sin^2(\theta)}$, which is non-negative on the range of $\arcsin$

4.4 The logistic function

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$$\sigma'(x) = \frac{e^{-x}}{(1 + e^{-x})^2} = \frac{1}{1 + e^{-x}} \frac{e^{-x}}{1 + e^{-x}} = \sigma(x) (1 - \sigma(x))$$

since $1 - \sigma(x) = e^{-x}/(1 + e^{-x})$

4.5 Least squares

- $f(\mathbf{w} + \epsilon \mathbf{u}) = \|(\mathbf{y} - \mathbf{X} \mathbf{w}) - \epsilon \mathbf{X} \mathbf{u}\|^2 = f(\mathbf{w}) - 2 \epsilon \mathbf{u}^\top \mathbf{X}^\top (\mathbf{y} - \mathbf{X} \mathbf{w}) + O(\epsilon^2)$
- So $\nabla f(\mathbf{w}) = -2 \mathbf{X}^\top (\mathbf{y} - \mathbf{X} \mathbf{w}) = 2 \mathbf{X}^\top (\mathbf{X} \mathbf{w} - \mathbf{y})$

4.6 A matrix derivative

- $L(\mathbf{W} + \epsilon \mathbf{U}) = \|(\mathbf{W} \mathbf{x} - \mathbf{y}) + \epsilon \mathbf{U} \mathbf{x}\|^2 = L(\mathbf{W}) + 2 \epsilon (\mathbf{W} \mathbf{x} - \mathbf{y})^\top \mathbf{U} \mathbf{x} + O(\epsilon^2)$
- $(\mathbf{W} \mathbf{x} - \mathbf{y})^\top \mathbf{U} \mathbf{x} = \text{tr}(\mathbf{U}^\top (\mathbf{W} \mathbf{x} - \mathbf{y}) \mathbf{x}^\top)$ by the same argument as the example above, so

$$\frac{\partial L}{\partial \mathbf{W}} = 2 (\mathbf{W} \mathbf{x} - \mathbf{y}) \mathbf{x}^\top$$